

Quiz 2

Total Marks: 30

17 October 2025

Part A: For Those Who Think in Intervals

Span Logic is a formal logic used for specifying properties of systems, by observing them over intervals of time $[b, e]$, where $b, e \in \mathbb{N}$. We assume that the system is observable at all discrete points of time $0, 1, 2, \dots, j$, where $j \in \mathbb{N}$.

Let $\mathbf{PVar}$ be a finite set of propositional variables representing observable aspects of system state (for example, whether a light is on, or whether a signal is active). A *valuation* assigns a truth value (either 0 or 1) to each variable:

$$\mathbf{VAL}(\mathbf{PVar}) = \mathbf{PVar} \rightarrow \{0, 1\}$$

A *behaviour* is a finite, non-empty sequence of such valuations, i.e, it is a word in $[\mathbf{VAL}(\mathbf{PVar})]^+$.

Example. Consider the variables $\{p, q\}$. The following table represents a behaviour:

Time step	0	1	2	3	4
p	1	0	1	1	0
q	0	0	0	0	1

In this example, p holds initially, while q becomes true only at the end. **Span Logic** allows us to write logical formulas that capture such temporal properties precisely.

Given a set of propositional variables $\mathbf{PVar}$, we call **Prop** the set of propositional formulae over $\mathbf{PVar}$. The syntax of any formula in **Prop** is same as the usual propositional logic. Then any well formed formula in **Span Logic** takes the following syntax:

$$\varphi := [P]^0 \mid [P] \mid \varphi \wedge \varphi \mid \neg \varphi \mid \varphi \bullet \varphi$$

where φ is a well formed formula (wff) in **Span Logic** and $P \in \mathbf{Prop}$. We denote the set of all wffs of Span Logic by $\mathcal{SL}$.

Let $\sigma \in [\mathbf{VAL}(\mathbf{PVar})]^+$ be a behaviour, and let $|\sigma|$ denote the length of behaviour σ . Let $\sigma(i)$ denote its behaviour at point i , $0 \leq i \leq |\sigma| - 1$, and let $\text{dom}(\sigma) = \{0, 1, \dots, |\sigma| - 1\}$. In the example above, $|\sigma| = 5$, $\sigma(0) = \{p \mapsto 1, q \mapsto 0\}, \dots, \sigma(4) = \{p \mapsto 0, q \mapsto 1\}$ and $\text{dom}(\sigma) = \{0, 1, 2, 3, 4\}$. Let $\sigma, i \models P$ denote that $P \in \mathbf{Prop}$ evaluates to true at position i in σ . This, in the above example, $\sigma, 0 \models p \wedge \neg q$, $\sigma, 1 \models \neg p \wedge \neg q$ and so on. The set of intervals in σ is given by $\text{Intv}(\sigma) = \{[b, e] \in \text{dom}(\sigma) \times \text{dom}(\sigma) \mid b \leq e\}$, where each interval identifies a subsequence of σ between positions b and e (both included).

For a chosen behaviour σ , an interval $[b, e] \in \text{Intv}(\sigma)$, with $b \leq e$, we inductively define the

satisfaction of a formula φ in Span Logic denoted $\sigma, [b, e] \models \varphi$ as follows:

$$\begin{aligned} \sigma, [b, e] &\models [P]^0 \text{ iff } b = e \text{ and } \sigma, b \models P \\ \sigma, [b, e] &\models [P] \text{ iff } b < e \text{ and } \sigma, i \models P \text{ for all } b < i < e \\ \sigma, [b, e] &\models \neg\varphi \text{ iff } \sigma, [b, e] \not\models \varphi \\ \sigma, [b, e] &\models \varphi_1 \wedge \varphi_2 \text{ iff } \sigma, [b, e] \models \varphi_1 \text{ and } \sigma, [b, e] \models \varphi_2 \\ \sigma, [b, e] &\models \varphi_1 \bullet \varphi_2 \text{ iff for some } m, b \leq m \leq e, \sigma, [b, m] \models \varphi_1 \text{ and } \sigma, [m, e] \models \varphi_2 \end{aligned}$$

A formula $\varphi \in \mathcal{SL}$ is **satisfiable** iff one can find a behaviour σ of some finite length such that $\sigma, [0, |\sigma| - 1] \models \varphi$. We also write $\sigma \models \varphi$ whenever $\sigma, [0, |\sigma| - 1] \models \varphi$.

Question 1. [6 marks] Write formulae in $\mathcal{SL}$ capturing each of the following requirements. That is, express each of the following requirements as a wff in $\mathcal{SL}$.

If any part of a specification is ambiguous, clearly state your assumption and write the formula. You will be awarded marks only if your assumption is reasonable.

- (a) [1 marks] A formula, **point** representing a *point interval*. (An interval $[b, e]$ where $b, e \in \mathbb{N}$ is said to be a *point interval* if $b = e$)
- (b) [1 marks] A formula, **unit** representing a *unit interval*. (An interval $[b, e]$ where $b, e \in \mathbb{N}$ is said to be a *unit interval* if $e - b = 1$)
- (c) [1 marks] A formula, $\llbracket P \rrbracket$ such that, for any word σ and interval $[b, e] \in \text{Intv}(\sigma)$, $\sigma, [b, e] \models \llbracket P \rrbracket$ if and only if $b < e$ and $\sigma, i \models P$ for all $b \leq i < e$.
- (d) [1 marks] Let φ be a wff in $\mathcal{SL}$ and σ be any behaviour. Capture the requirement “ φ evaluates to true for all $[b, e] \in \text{Intv}(\sigma)$ ”.
- (e) [2 marks] Consider $\text{PVar} = \{p, q, r, s\}$. Write a formula, $\varphi \in \mathcal{SL}$ which holds over a behaviour σ such that

- $\sigma, 0 \models p$,
- $\sigma, i \models \neg p$ for $0 < i \leq |\sigma| - 1$,
- $\sigma, |\sigma| - 1 \models q$,
- $\sigma, i \models \neg q$ for $0 \leq i < |\sigma| - 1$ and
- $\sigma, i \models r \oplus s$ for $0 < i < |\sigma| - 1$.

Solution

Here, $\top$ denotes a tautology in propositional logic, or $\top$ can be represented as $p \vee \neg p$. Similarly, $\perp$ denotes a contradiction, or can be represented as $p \wedge \neg p$.

- (a) $\llbracket \top \rrbracket^0$
- (b) $\llbracket \perp \rrbracket$
- (c) $\llbracket p \rrbracket^0 \bullet \llbracket p \rrbracket$ or another valid solution is $\llbracket p \rrbracket \wedge (\llbracket p \rrbracket^0 \bullet \llbracket \top \rrbracket)$
- (d) $\neg(\text{true} \bullet \neg\varphi \bullet \text{true})$ where *true* is defined as $(\text{point} \vee \neg\text{point})$. The idea is to write a

formula which encodes $\neg\varphi$ being true for some interval in σ then negating it gives the solution to the question

$$(e) \ [p \wedge \neg q]^0 \bullet [\neg p \wedge \neg q \wedge (r \wedge \neg s) \vee (\neg r \wedge s)] \bullet [\neg p \wedge q]^0$$

Rubrics

Full marks awarded even if solution has minor modifications or redundant conditions which still ensure correctness. Note that 0.5 marks deducted if $\vee$ is used in the $\mathcal{SL}$ formula (but can be used in the propositional formula)

We extend **Span Logic** to handle more interesting formula. We call this **Extended Span Logic** (denoted as $e\mathcal{SL}$). The syntax of $e\mathcal{SL}$ is given below:

$$\psi := [P]^0 \mid [P] \mid \psi \wedge \psi \mid \neg\psi \mid \psi \bullet \psi \mid \text{len} \sim c \mid \int P \sim c$$

where $\psi \in e\mathcal{SL}$, $c \in \mathbb{N}$, $\sim \in \{>, =\}$ and $P \in \text{Prop}$.

$\text{len} \sim c$ and $\int P \sim c$ are called measurements. len stands for the length of the interval chosen. $\int P$ is the number of times P holds good in a chosen interval $[b, e]$, and this number is compared with c in evaluation of $\int P \sim c$. For a given behaviour σ over $[b, e]$ where $b < e$, define

$$\int_b^e P = \sum_{i=b}^{e-1} x_i$$

where $x_i = 1$ if $\sigma, i \models P$, and $x_i = 0$ otherwise.

$$\begin{aligned} \sigma, [b, e] \models \text{len} \sim c & \text{ iff } e - b \sim c \\ \sigma, [b, e] \models \int P \sim c & \text{ iff } \int_b^e P \sim c \end{aligned}$$

In our example, we thus have $\sigma, [0, 3] \models (\text{len} = 3)$, $\sigma, [0, 3] \models (\int p = 2)$.

A formula $\varphi \in e\mathcal{SL}$ is **satisfiable** iff one can find a behaviour σ of some finite length such that $\sigma, [0, |\sigma| - 1] \models \varphi$. We also write $\sigma \models \varphi$ whenever $\sigma, [0, |\sigma| - 1] \models \varphi$.

Question 2. [8 marks] Compare the expressiveness of $e\mathcal{SL}$ and $\mathcal{SL}$. In other words, for every $\psi \in e\mathcal{SL}$, does there exist a $\varphi \in \mathcal{SL}$ such that $\sigma \models \varphi \iff \sigma \models \psi$ for any behaviour σ ? Give a complete formal proof.

Solution

The expressiveness of $e\mathcal{SL}$ and $\mathcal{SL}$ is the same. We prove this by giving transformation of each syntactic element in both logics in the other logic

1. $e\mathcal{SL} \rightarrow \mathcal{SL}$ Let $\alpha : e\mathcal{SL} \rightarrow \mathcal{SL}$

$$\begin{aligned}
\alpha[\lceil P \rceil^0] &= \lceil P \rceil^0 \\
\alpha[\lceil P \rceil] &= \lceil P \rceil \\
\alpha[\psi_1 \wedge \psi_2] &= \alpha[\psi_1] \wedge \alpha[\psi_2] \\
\alpha[\neg \psi] &= \neg \alpha[\psi] \\
\alpha[\psi_1 \bullet \psi_2] &= \alpha[\psi_1] \bullet \alpha[\psi_2] \\
\alpha[\text{len} = c] &= \begin{cases} \lceil \top \rceil^0 & \text{if } c = 0 \\ \underbrace{\lceil \perp \rceil \dots \lceil \perp \rceil}_{c \text{ times}} & \text{otherwise} \end{cases} \\
\alpha[\text{len} > c + 1] &= \alpha[\text{len} = c] \bullet \lceil \top \rceil \\
\alpha[\int P = 0] &= \neg \lceil \top \rceil^0 \wedge \neg \lceil \neg P \rceil \\
\alpha[\int P = 1] &= \alpha[\int P = 0] \bullet (\lceil P \rceil \wedge \lceil \perp \rceil) \bullet \alpha[\int P = 0] \\
\alpha[\int P = c + 1] &= \alpha[\int P = 1] \bullet \alpha[\int P = c] \\
\alpha[\int P > c] &= \alpha[\int P = c] \bullet (\neg(\lceil \top \rceil^0 \wedge \lceil \top \rceil) \bullet \lceil P \rceil \bullet \neg(\lceil \top \rceil^0 \wedge \lceil \top \rceil))
\end{aligned}$$

2. Since $e\mathcal{SL}$ is an extension of $\mathcal{SL}$, the former is at least as expressive as the latter. So, $e\mathcal{SL}$ and $\mathcal{SL}$ are equally expressive.

Rubrics

- No marks have been awarded for just stating the equivalence of $e\mathcal{SL}$ and $\mathcal{SL}$ without giving the transformations for at least one of $\text{len} \sim c$ or $\int P \sim c$
- Full marks have been awarded only if the transformations for $\text{len} \sim c$, $\int P \sim c$ have been given **and** the fact that other formulae are present in both logics is explicitly mentioned. If the latter is missing, 1 mark has been deducted.
- No marks have been deducted for not mentioning that the $e\mathcal{SL}$ is at least as expressive as $\mathcal{SL}$
- In $\text{len} \sim c$,
 1. $\sim \mapsto =$
0.5 mark has been given for mentioning $c = 0$ case, 1.5 marks have been given for the $c \geq 1$ case
 2. $\sim \mapsto >$
1 mark for a fully correct formula, 0.5 mark deducted for missing corner cases
- In $\int P \sim c$,
 1. $\sim \mapsto =$
For mentioning the $c = 0, 1$ cases 0.5 mark each, 1 mark has been given for the $c > 1$ case
 2. $\sim \mapsto >$
2 marks for a fully correct formula, 0.5 marks deducted for missing any corner

case.

Question 3. [10 marks] Show that for every $\psi \in e\mathcal{SL}$ over a set of propositional variables PVar , we can construct an MSO formula Φ such that for all behaviors $\sigma \in [\text{VAL}(\text{PVar})]^+$,

$$\sigma \models \psi \iff \sigma \models \Phi$$

Solution

Define a transformation map, $\Omega : \mathcal{SL} \mapsto \text{MSO}$ as follows:

$$\begin{aligned}\Omega(\lceil P \rceil^0) &\triangleq (x = y \wedge P'(x)) \\ \Omega(\lceil P \rceil) &\triangleq (x < y \wedge \forall z ((x < z \wedge z < y) \rightarrow P'(z))) \\ \Omega(\varphi_1 \wedge \varphi_2) &\triangleq \Omega(\varphi_1) \wedge \Omega(\varphi_2) \\ \Omega(\neg \varphi) &\triangleq ((x < y \vee x = y) \wedge \neg \Omega(\varphi)) \\ \Omega(\varphi_1 \bullet \varphi_2) &\triangleq \exists z (\Omega(\varphi_1)[z/y] \wedge \Omega(\varphi_2)[z/x])\end{aligned}$$

where, $P \in \text{Prop}$ and $\varphi_1, \varphi_2, \varphi \in \mathcal{SL}$. For a $P \in \text{Prop}$, we introduce a second order variable P' which is true exactly at positions where P holds true.

From Qn 2, we know that for every formula in $e\mathcal{SL}$, there is an equivalent formula in $\mathcal{SL}$. For $\psi \in e\mathcal{SL}$, let $\Lambda(\psi) \in \mathcal{SL}$ denote the equivalent $\mathcal{SL}$ formula. Then the required Φ is given by

$$\Phi = \exists x \exists y (\text{first}(x) \wedge \text{last}(y) \wedge \Omega(\Lambda(\psi)))$$

where **first** and **last** are predicates, defined as usual indicating whether a position is the first or last (resp.).

Remarks.

- The correctness of the construction follows directly from the semantics (go back and check the way semantics is defined and you will realize the striking similarity :))
- Observe that the alphabet of words when we consider the MSO formula is $\{0, 1\}^{\text{PVar}}$ and not $\{0, 1\}$. Thus given a letter in this alphabet, we can say whether a $P \in \text{Prop}$ is true or false. This is how we interpret if $P'(x)$ is true at a position given a word.
- Notice that we didn't define $\Omega(\neg \varphi) \triangleq \neg \Omega(\varphi)$, because any behaviour is **non-empty** sequence of valuations. Thus, our transformed MSO formula should not accept empty word.
- You can also construct a DFA from ψ following the semantics; however you should work with correct alphabet.

Rubrics

The expected rubrics was to give 2 marks per transformation with a short justification on the correctness (as correctness proof is just re-writing the semantics). However only less than 20 people attempted this question. Additionally, there were quite less reasonable

attempts. Only 2 people received *non-zero* marks (as of now). The marks are awarded as per merit.

PS: No marks for just writing the transformation for $\varphi_1 \wedge \varphi_2$ or anything trivial.

Part B: Because you learnt Ground Resolution

Question 4. [6 marks] Consider the following set of FOL formulae.

$$F_1 = \exists x (P(x) \wedge \neg P(f(x)))$$

$$F_2 = \forall x (\neg P(x) \vee P(f^m(x)))$$

$$F_3 = \forall x (P(x) \vee \neg P(f^n(x)))$$

where $f^k(x)$ refers to the term $\underbrace{f(\cdots(f(x))\cdots)}_{k \text{ times}}$. For example, $f^3(x)$ denotes $f(f(f(x)))$.

- Let $m = 3$ and $n = 4$. Is $F_1 \wedge F_2 \wedge F_3$ is satisfiable? If it is satisfiable, exhibit a model under which it is true. Otherwise, show that it is unsatisfiable using ground resolution.
- Let $m = 3$ and $n = 6$. Is $F_1 \wedge F_2 \wedge F_3$ is satisfiable? If it is satisfiable, exhibit a model under which it is true. Otherwise, show that it is unsatisfiable using ground resolution.

Solution

- For $m = 3$ and $n = 4$, $F_1 \wedge F_2 \wedge F_3$ is not satisfiable. Skolemizing F_1, F_2, F_3 gives,

$$G_1 = P(c) \wedge \neg P(f(c))$$

$$G_2 = \forall x (\neg P(x) \vee P(f^3(x)))$$

$$G_3 = \forall x (P(x) \vee \neg P(f^4(x)))$$

To carry out resolution we consider the following ground instances:

$$\{P(c)\} \quad (\text{from } G_1)$$

$$\{\neg P(f(c))\} \quad (\text{from } G_1)$$

$$\{\neg P(c), P(f^3(c))\} \quad (\text{from } G_2)$$

$$\{\neg P(f^2(c)), P(f^5(c))\} \quad (\text{from } G_2)$$

$$\{\neg P(f^3(c)), P(f^6(c))\} \quad (\text{from } G_2)$$

$$\{P(f(c)), \neg P(f^5(c))\} \quad (\text{from } G_3)$$

$$\{P(f^2(c)), \neg P(f^6(c))\} \quad (\text{from } G_3)$$

Then following propositional resolution proof derives $\emptyset$ from the above subset of ground

instances:

1. $\{P(c)\}$
2. $\{\neg P(c), P(f^3(c))\}$
3. $\{P(f^3(c))\}$ Resolution on 1, 2
4. $\{\neg P(f^3(c)), P(f^6(c))\}$
5. $\{P(f^6(c))\}$ Resolution on 3, 4
6. $\{P(f^2(c)), \neg P(f^6(c))\}$
7. $\{P(f^2(c))\}$ Resolution on 5, 6
8. $\{\neg P(f^2(c)), P(f^5(c))\}$
9. $\{P(f^5(c))\}$ Resolution on 7, 8
10. $\{P(f(c)), \neg P(f^5(c))\}$
11. $\{P(f(c))\}$ Resolution on 9, 10
12. $\{\neg P(f(c))\}$
13. $\emptyset$ Resolution on 11, 12

(b) Yes. Define model M with $\mathcal{U}_M = \{a, b, c\}$, f_M interpreted as

$$f_M(a) = b, f_M(b) = c, f_M(c) = a$$

and, $P_M = \{a\}$

Since $\forall x, f_M^3(x) = f_M^6(x) = x$, F_2 and F_3 are true. For $x = a$, F_1 is true.

Rubrics

- (a)
 - [3 marks] if contradiction is reached using ground resolution correctly
 - 1 mark deducted if steps are skipped without proof (by induction)
 - *Note:* No marks if contradiction is not reached
- (b)
 - [2 marks] Giving a model which satisfies the formula.
 - [1 mark] Explaining how the model satisfies the formula.
 - *Note:* No marks for *almost correct* models